

9(a)

any **eight** from:

- 1 ground substance / stroma ;
- 2 for, light independent stage / Calvin cycle ;
- 3 contains enzymes / named enzyme e.g. rubisco ;
- 4 also, sugars / lipids / starch / ribosomes / DNA ;
- 5 internal membrane system ;
- 6 for, light dependent stage ;
- 7 fluid-filled sacs / thylakoids ;
- 8 grana are stacks of thylakoids ;
- 9 (grana) hold (photosynthetic) pigments ;
- 10 (grana) have large surface area for (maximum) light absorption ;
- 11 (pigments are arranged in), light harvesting clusters / photosystems ;
- 12 primary pigment / reaction centre / chlorophyll a, surrounded by accessory pigments ;
- 13 (accessory pigments) pass energy to, primary pigment / reaction centre / chlorophyll a ;
- 14 different photosystems absorb light at different wavelengths ;
- 15 membranes hold, ATP synthase / electron carriers ;
- 16 for, photophosphorylation / chemiosmosis ;

Question	Answer	Marks
9(b)	<i>any seven from:</i> 1 grind leaf ; 2 use, solvent / named solvent ; 3 leaf extract contains mixture of pigments ; 4 <i>ref. to</i> concentrate extract ; 5 further detail ; e.g. pencil line drawn / extract placed on chromatography paper / extract placed on thin layer chromatography plate 6 paper placed (vertically) in jar of (different) solvent ; 7 solvent rises up paper ; 8 each pigment travels at different speed ; 9 pigments separated as they ascend ; 10 distance moved by each pigment is unique ; 11 R_f value ; 12 two dimensional chromatography ; 13 better separation of pigments ;	7